

Table of Contents

AlphaX POS — Administrator Manual

For: System administrators, IT staff, business owners doing setup **Version:** 15.5.13 **Last updated:** May 2026

What's in this manual

1. Overview: what AlphaX POS includes
 2. Architecture: three apps, what runs where
 3. Pre-deployment checklist
 4. Day 1: install the apps on your bench
 5. Day 2: company, branch, outlet, terminal hierarchy
 6. Day 3: POS Profile and payment methods
 7. Day 4: items, taxes, pricing
 8. Day 5: users, roles, manager PINs
 9. Day 6: ZATCA Phase 2 setup
 10. Day 7: hardware bridge — printers, drawers, scanners
 11. Domain packs: configuring per-vertical features
 12. Loyalty, offers, pricing rules
 13. Pharmacy domain: drugs, prescriptions, controlled substances
 14. Routine administration tasks
 15. Backup and disaster recovery
 16. Troubleshooting
 17. Glossary
-

Overview: what AlphaX POS includes

✓ **Working** unless marked otherwise.

AlphaX POS is a Frappe v15 app suite for retail, restaurants, and pharmacies. It's designed for the Saudi Arabian market with built-in ZATCA Phase 2 e-invoicing.

What's in the box

- **POS register UI** for cashier transactions (/app/alphax-pos-classic)
- **52 doctypes** covering outlets, terminals, profiles, orders, shifts, cash moves, theme settings, drug master, prescriptions, controlled substance log, loyalty, central kitchen, ZATCA processing log, and more

- **Workspace** (/app/alphax-pos-hub) with shortcuts to all the major setup and operational areas
- **Manager PIN system** (AlphaX POS Manager PIN) with bcrypt hashing, exponential lockout, full audit log
- **POS Profile integration** with standard ERPNext POS Profile
- **Domain packs** to enable/disable features per vertical
- **Hardware bridge** (separate Python daemon) for thermal printers, cash drawers, scales, card terminals — runs on each cashier PC
- **ZATCA Phase 2** integration via the alphax-zatca companion app (forked from ERPGulf)

What's optional

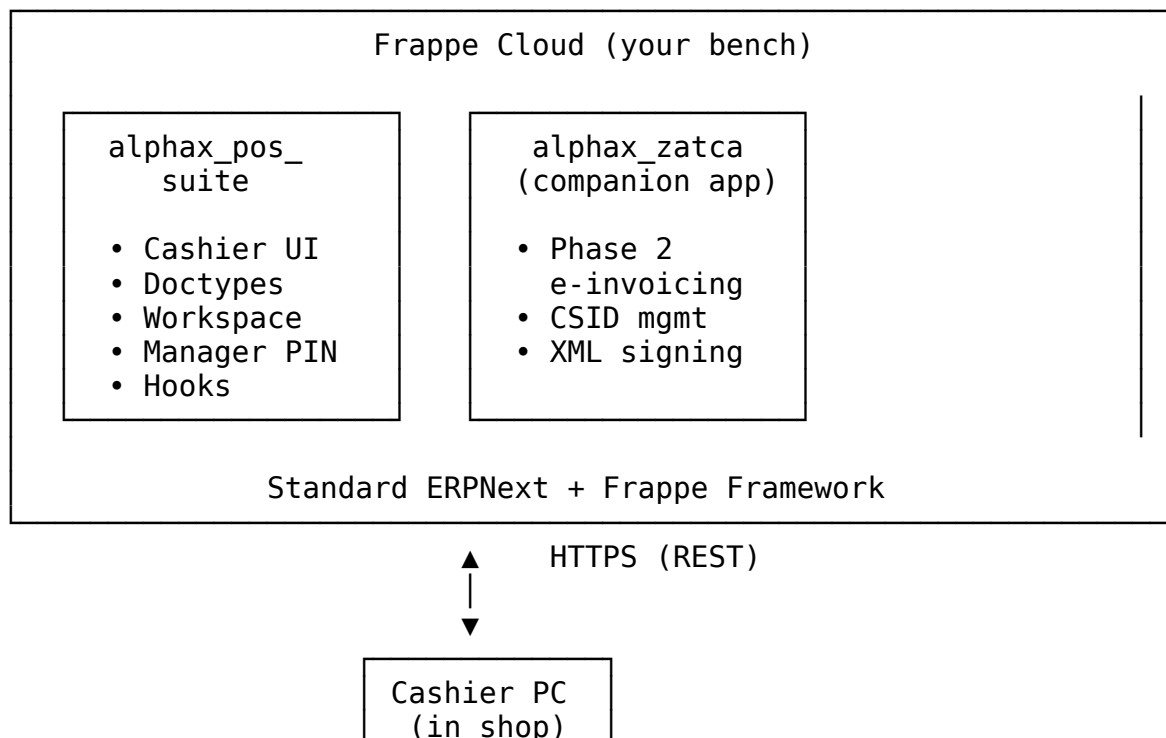
-  **Vue cashier** at /app/alphax-pos-v2 — a richer single-page-app cashier (in Beta; the jQuery alphax-pos-classic is the production default for v15.5.13)
-  **Floor Designer, KDS** (Kitchen Display System), **Profitability Dashboard, APOS** (alternative cashier) — beta features useful for restaurants

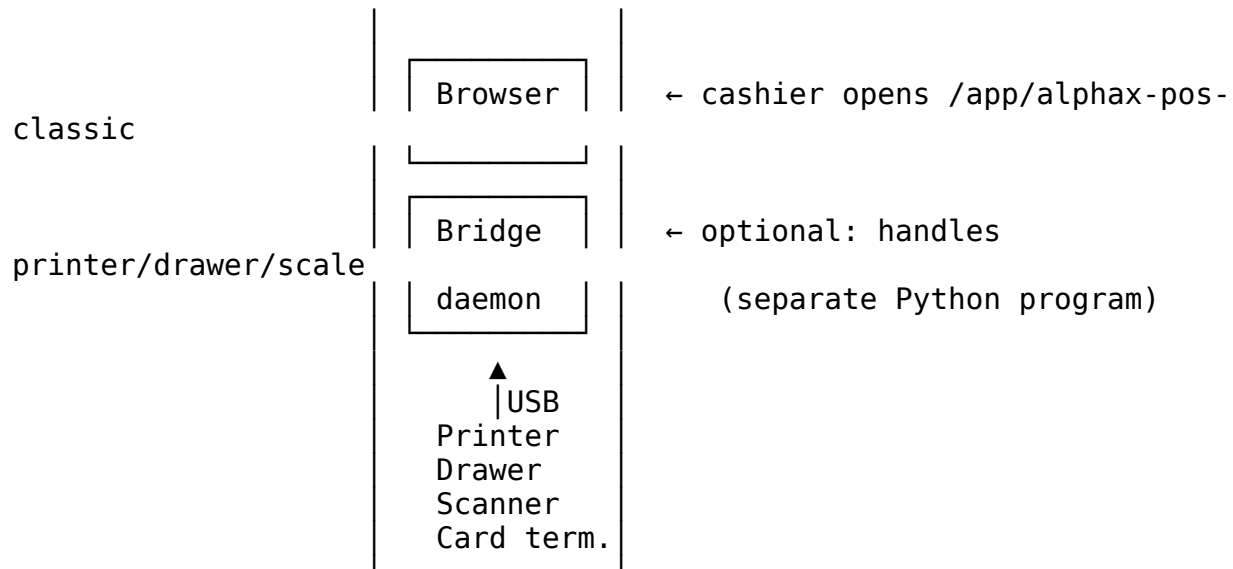
Status conventions

✓ **Working** — production-ready  **Beta** — works, but the API or UI may shift in a near release  **In Development** — listed for completeness; do not deploy yet

Architecture: three apps, what runs where

The three components





Why three apps and not one

- **alphax_pos_suite** — the main app. Lives in the GitHub repo <https://github.com/jamunachi08/alphax-pos-core-v15.5.1>. Contains the cashier, doctypes, workspace, security module.
- **alphax_zatca** — a fork of ERPGulf's `zatca_erp_gulf` app. Kept separate because (a) it changes independently of the POS, (b) some customers use only the POS without ZATCA (non-KSA shops), (c) the upstream fork retains GPL-3 licensing.
- **alphax-pos-bridge** — a standalone Python daemon, NOT a Frappe app. Runs on each cashier PC. Talks to USB hardware. Communicates with the browser via WebSocket on `ws://localhost:9876`.

Data flow at a sale

1. Cashier scans barcode → browser POSTs to Frappe (Sales Invoice)
 2. Frappe creates the invoice → triggers `on_submit` hooks
 3. `alphax_pos_suite` hooks update inventory, loyalty, audit log
 4. `alphax_zatca` hook signs the invoice XML and submits to ZATCA portal
 5. Frappe responds to browser
 6. Browser tells the bridge daemon to print receipt + open drawer
 7. Bridge daemon talks to printer/drawer over USB
-

Pre-deployment checklist

Before you start, confirm you have:

Frappe Cloud account

- ☐ A Frappe Cloud account (frappecloud.com)
- ☐ A bench provisioned for this customer (Singapore or AWS-Bahrain region recommended for KSA latency)

- ☐ Your bench's URL (e.g., `customername.k.frappe.cloud`)
- ☐ System Administrator credentials for the site

Apps and source code

- ☐ Access to the GitHub repository: `https://github.com/jamunachi08/alphax-pos-core-v15.5.1`
- ☐ Same for the ZATCA fork (separate repo, ask the project owner)
- ☐ If self-hosted, the bench server has SSH access

Customer information

- ☐ Customer's full legal name (for ZATCA registration)
- ☐ VAT number (15 digits)
- ☐ CR (Commercial Registration) number
- ☐ Branch addresses (each branch needs to be set up)
- ☐ Number of cashier stations per branch
- ☐ Hardware list: printer make/model, scanner brand, card terminal provider, cash drawer model

ZATCA-specific (if KSA customer)

- ☐ Customer is registered on the [Fatoora portal](#)
- ☐ Customer has agreed to provide the OTP from Fatoora when needed
- ☐ You know which production environment ZATCA uses for them (Phase 2 generation, post-2024)

Hardware (per cashier PC)

- ☐ Windows 10/11 PC, Linux Ubuntu 20+, or macOS 12+
 - ☐ Modern browser (Chrome, Edge, Firefox)
 - ☐ Stable internet connection
 - ☐ USB ports for hardware
 - ☐ Optionally: a label-maker for printing PC stickers (so manager knows "this PC = Terminal A07")
-

Day 1: install the apps on your bench

Working

1.1 Add the GitHub repo to your bench

In Frappe Cloud dashboard:

1. Go to **Bench** → **Apps** → **Add App**
2. Source: **From GitHub Repository**
3. URL: `https://github.com/jamunachi08/alphax-pos-core-v15.5.1`
4. Branch: `main` (or whichever your team uses)
5. Click **Add**

Frappe Cloud's deploy pipeline will pull, build, and prepare the app.

1.2 Install on your site

After the deploy succeeds:

1. Go to **Site** → **Apps**
2. Click **Install** next to `alphax_pos_suite`
3. Wait for the green check (~1-2 minutes)

1.3 (Optional) Install the ZATCA companion app

If your customer is in KSA and needs Phase 2 e-invoicing:

1. Repeat 1.1 with the ZATCA repo URL
2. Install on the same site

1.4 Verify installation

Open `/app/alphax-pos-hub` in a browser. You should see the workspace with 9 module cards: Operations, Setup & Configuration, Catalog & Pricing, Sales & Customers, Pharmacy, Compliance & Tax, Reports & Analysis, Loyalty & Offers, Data Import & Settings.

If you see "Not Found" or the workspace is empty, see the troubleshooting section.

1.5 Verify versions

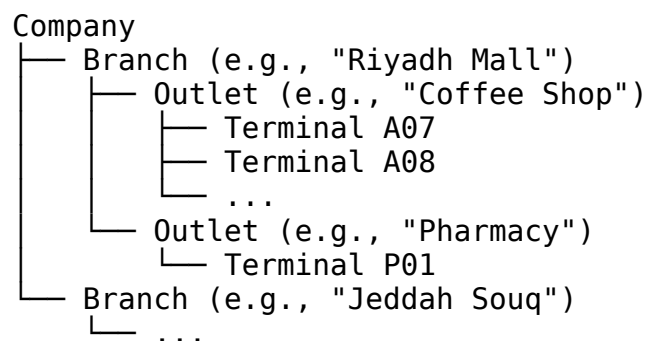
Click anywhere → **Help** → **About** in the desk. The Installed Apps list should show:

-  **alphax_pos_suite: v15.5.13** (or latest)
 -  **alphax_zatca: v3.0.x** (if installed)
-

Day 2: company, branch, outlet, terminal hierarchy

Working

This is the core data hierarchy. Get it right before doing anything else.



2.1 Create the company

If new install, ERPNext will already prompt you. If not:

1. Go to `/app/company/new`
2. Required: Company Name, Abbreviation, Default Currency (SAR for KSA), Country (Saudi Arabia)
3. ZATCA-required: Tax ID (15-digit VAT number), Company Registration Number, Address (with district, city, postal code, country)
4. Save

 **Don't skip the address.** ZATCA Phase 2 requires the seller's address on every invoice. Missing fields cause invoice submission to fail.

2.2 Create branches

For each physical store location:

1. Go to `/app/branch/new`
2. Branch name: human-readable (e.g., "Riyadh Mall — Branch 02")
3. Save

2.3 Create outlets

For each business unit within a branch:

1. Go to `/app/alphax-pos-outlet/new`
2. Outlet Name: human-readable (e.g., "Coffee Shop")
3. Branch: link to the parent branch
4. Domain Pack (optional): pick the vertical (Restaurant, Retail, Pharmacy, Electronics, Generic) — affects which features the cashier shows
5. Default Warehouse: where stock for this outlet lives (often a Branch-level warehouse)
6. Default Cost Center
7. Save

 **Outlets are the operational unit** — most reports and permissions are by outlet, not branch. Create one outlet per business unit even if the branch only has one shop.

2.4 Create terminals

For each physical cashier station:

1. Go to `/app/alphax-pos-terminal/new`
2. Terminal Name: short code (e.g., "A07", "P01")
3. POS Outlet: link to the parent outlet
4. (Optional) Default POS Profile: a POS Profile to suggest
5. (Optional) Hardware tags: which printer, which scanner, etc.
6. Save

💡 **Use a labeling system for terminal names** that makes physical sense. “A07” = “Aisle A, station 07”. Or “RM-CS-01” = “Riyadh Mall, Coffee Shop, station 01”. Whatever you pick, label the actual PCs with stickers so managers know which terminal to bind during setup.

2.5 Verify the hierarchy

Open `/app/alphax-pos-terminal` and look at the list. You should see each terminal with its outlet visible. Click into one — the breadcrumb should show you Branch — Outlet — Terminal.

Day 3: POS Profile and payment methods

✓ Working

POS Profiles are standard ERPNext doctypes that define a cashier’s operational permissions. AlphaX POS uses these directly.

3.1 Create payment methods (Mode of Payment)

Before profiles, make sure you have payment methods:

1. Go to `/app/mode-of-payment`
2. Standard ones: Cash, Mada (KSA debit), Visa, Mastercard, Apple Pay, Bank Transfer, Credit Note, Loyalty Points
3. Each has an “Account” field — link to the appropriate cash or bank account in your chart of accounts
4. Mark which are “POS”-enabled

3.2 Create a POS Profile

For each role/outlet combination:

1. Go to `/app/pos-profile/new`
2. Name: human-readable (e.g., “Coffee Shop Cashier”)
3. Company, Customer (default walk-in), Currency
4. **Warehouse:** where stock comes from for this profile’s sales
5. **Mode of Payment table:** list all accepted payment methods
6. **Applicable for Users:** list specific cashiers OR leave empty for “all”
7. **Item Groups:** filter what items show up
8. **Taxes and Charges Template:** link to a Sales Tax template
9. **Letter Head, Print Heading:** for receipts
10. **Update Stock** (check — usually yes)
11. **Apply Discount on:** Grand Total or Net Total
12. Save

3.3 Link the profile to your AlphaX outlet

After saving, also link the POS Profile to the AlphaX POS Outlet:

1. Open the AlphaX POS Outlet record
2. Set “Default POS Profile” if there’s only one for this outlet
3. Or set per-cashier in their User record (if your shop assigns by user)

3.4 Test it

The cashier won’t fully work until users are assigned and PINs are set, but you can verify the data setup by going to `/app/alphax-pos-classic` as System Administrator. The “Station Not Configured” card should appear (because no terminal is bound to your admin’s PC). Don’t bind it yet — that’s for cashier PCs, not yours.

Day 4: items, taxes, pricing

Working

4.1 Item groups

Create item groups that match how your customer thinks about their products:

```
All Item Groups
├── Beverages
│   ├── Hot Drinks
│   └── Cold Drinks
├── Food
│   ├── Sandwiches
│   └── Desserts
├── Pharmacy (separate vertical)
│   ├── OTC
│   └── Prescription
└── Services
```

4.2 Items

Create each sellable product:

1. Go to `/app/item/new`
2. Item Code: short, scannable (e.g., “WAT-001”)
3. Item Name: human-readable
4. Item Group
5. Default Unit of Measure
6. **Maintain Stock:** yes for retail/pharmacy, no for services
7. **Is Sales Item:** yes
8. **Standard Selling Rate** in the Item Defaults table
9. Save

 **For barcode-driven workflows**, the Item Code IS the barcode. Or add separate barcodes via the Item's "Barcodes" child table.

4.3 Item taxes

For VAT (KSA) and similar:

1. Go to /app/sales-taxes-and-charges-template
2. Create "KSA Standard 15% VAT" template
3. Add row: Account Head = "VAT 15% - Output", Rate = 15
4. Mark the "Inclusive" checkbox if your prices include VAT
5. Save

Link this template on POS Profiles → Taxes section.

4.4 Inclusive vs exclusive VAT

KSA shops typically display prices INCLUSIVE of VAT (the customer sees the total they pay). Configure:

- POS Profile → "Apply Discount on" → Net Total
- Sales Tax Template → mark "Included in Print Rate"
- Item rate = price-with-VAT (e.g., 11.50 SAR for a 10-SAR-net coffee)

The system breaks out VAT on the receipt automatically.

4.5 Item pricing rules

For "Buy 2 get 1 free", "10% off Tuesdays", loyalty discounts:

1. Go to /app/pricing-rule/new
2. Filter: by item, item group, customer group, or all
3. Apply on: Discount Percentage / Discount Amount / Free Item
4. Conditions: Min/Max quantity, Date range, Day of week
5. Save

4.6 Test with a dummy item

Open the cashier (you'll need to bind your test PC first). Scan or type the item code. Confirm the price, taxes, and any discounts are applied as expected.

Day 5: users, roles, manager PINs

 **Working**

5.1 Create the AlphaX POS Manager role (if not present)

The role is auto-created on app install, but verify:

1. Go to /app/role
2. Search for “AlphaX POS Manager”
3. If missing, create with:
 - Name: AlphaX POS Manager
 - Description: “Authorizes register-side privileged actions (returns, discounts, station setup, voids)”
4. Save

5.2 Create cashier users

For each cashier:

1. Go to /app/user/new
2. First Name, Last Name, Email (will be username)
3. Set a temporary password OR send password reset link
4. **Roles** tab: assign “Sales User” + “POS User” + (optional) “Stock User”
5. **Restrict by User Permissions**: optionally restrict by company, warehouse
6. Save

5.3 Create manager users

For each manager:

Same as cashiers, plus: 1. **Roles** tab: ALSO assign “AlphaX POS Manager” 2. Don’t grant System Manager unless they truly need full admin

5.4 Set manager PINs

This is currently CLI-only — there’s no admin UI for it yet.

In Frappe Cloud: 1. Site → Console (in the bench dashboard) 2. Run:

```
import frappe
frappe.call(

    'alphax_pos_suite.alphax_pos_suite.security.manager_pin.set_manager_pin',
    user='khalid@example.com',
    pin='8472'
)
```

Replace the email and PIN. The PIN must be 4-6 digits.

The system: - Verifies the user has a manager role - Hashes the PIN with bcrypt (12 rounds) - Saves to AlphaX POS Manager PIN doctype - Resets any prior logout state - Logs the action in the Authorization Log

💡 **Tell the manager their PIN privately.** Don’t email it. Don’t WhatsApp it. Walk over and tell them, or use a password manager share if your team has one.

5.5 Test PIN authentication

1. Open /app/alphax-pos-classic on a non-bound PC (the test bench admin's own browser is fine)
2. Click "Manager Setup"
3. Type the manager's username + PIN
4. Verify the green "Authorized" message appears
5. Pick any terminal to bind (you can reset right after)
6. Confirm the cashier UI loads

5.6 Reset a forgotten PIN

If a manager forgets:

```
import frappe
frappe.call(

'alphax_pos_suite.alphax_pos_suite.security.manager_pin.set_manager_pin',
    user='khalid@example.com',
    pin='9999' # NEW pin
)
```

This resets the PIN AND clears any logout state.

5.7 Reset a locked-out manager

If logout was triggered without resetting the PIN itself:

```
import frappe
frappe.call(

'alphax_pos_suite.alphax_pos_suite.security.manager_pin.reset_manager_lockout',
    user='khalid@example.com'
)
```

Day 6: ZATCA Phase 2 setup

✅ **Working** for the integration code 🚧 **In Development** — full guided onboarding UI (manual steps for now)

⚠️ **This section is critical for KSA customers.** Mistakes mean invoices don't submit, leading to compliance violations.

6.1 Confirm ZATCA registration

Before configuring the system, your customer must:

1. Be registered on [Fatoora portal](#)

2. Have a valid VAT certificate
3. Be aware they're moving to Phase 2 (Integration / Generation phase)

If they're not yet registered, that's a tax-consultant task — not a software task. Don't proceed without it.

6.2 Generate device certificates

Each cashier device that submits invoices needs its own CSID (Cryptographic Stamp Identifier). The flow:

1. Cashier admin opens the AlphaX ZATCA app's onboarding page
2. Enter OTP from the Fatoora portal
3. The app generates a private key + CSR
4. App submits CSR to ZATCA
5. ZATCA returns the device certificate
6. Certificate is stored on the device (or in the bench, depending on your deployment model)

 **Manual step for v15.5.13:** The full guided UI is in development. For now, follow ERPGulf's documentation in the `alphax_zatca` repo for the device-onboarding flow. The fork retains their tooling.

6.3 Production vs sandbox

ZATCA has separate sandbox and production environments. Configure:

1. Go to `/app/alphax-zatca-settings` (or equivalent in the fork)
2. Set "Mode" = Sandbox while testing
3. Switch to Production only after successful sandbox tests

6.4 Verify e-invoicing on a test sale

1. Make a test sale on a cashier PC
2. Check `/app/alphax-pos-processing-log` (the workspace shortcut "ZATCA Status")
3. The new invoice should appear with status "Submitted" or "Pending"
4. After a few seconds, "Submitted" should become "Cleared"
5. The QR code on the receipt should be a valid Phase 2 ZATCA QR

If status stays "Failed", drill into the row for the error message. Most common causes: - Device certificate not provisioned for this terminal - VAT number missing on customer - Inclusive VAT misconfigured - Time desync between bench and ZATCA (rare; ZATCA is strict on timestamps)

Day 7: hardware bridge — printers, drawers, scanners

 **Working** for browser-only flows (no special hardware needed)  **Beta** for the bridge daemon

7.1 What works without the bridge

Out of the box, with a standard browser and a USB receipt printer configured at the OS level, you can:

- Print receipts via the browser's print dialog
- Use a USB barcode scanner (acts as a keyboard, no setup needed)
- Capture cash payments (no drawer hardware needed for amount entry)

For many small shops, this is enough. Skip to Day 8.

7.2 What requires the bridge

If you need:

- Auto-cut receipts (no print dialog)
- Cash drawer that opens automatically after a sale
- Weight scale integrated with item entry
- Card terminal that exchanges payment data with the POS automatically
- Direct ESC/POS commands to specific printer models

...you need the AlphaX POS Bridge daemon running on each cashier PC.

7.3 Installing the bridge

The bridge is a separate Python program in its own GitHub repo.

 **Status:** The bridge codebase exists with installer scaffolding (PyInstaller, Inno Setup for Windows, .deb for Linux), but the end-to-end installer has not been certified for production. For v15.5.13 deployments, treat the bridge as Beta.

7.4 Bridge → browser communication

The bridge listens on `ws://localhost:9876`. The browser connects to it for hardware actions:

Browser ↔`WebSocket`→ Bridge daemon ↔`USB`→ Hardware

If the bridge is offline, the cashier still works — it just can't auto-print or open the drawer.

7.5 Testing the bridge

After install:

1. Start the bridge daemon (`alphax-bridge` command)
2. In the cashier, click any test buttons (planned for `/app/alphax-pos-hardware-test`)
3. Verify printer prints, drawer opens, scanner reads

 **The hardware test page is in development.** For v15.5.13, test by completing a real sale and verifying receipt + drawer behavior.

Domain packs: configuring per-vertical features

✓ Working

A “domain pack” is a configuration that turns on/off features for a vertical. You set the domain pack per Outlet.

Available packs

- **Restaurant** — Floor Plan, KDS, Modifiers, Service Charge, Tips
- **Retail** — Standard cart with stock check
- **Pharmacy** — Drug Master, Prescriptions, Controlled Substance Log, Drug Interaction Rules
- **Electronics** — Serial Number capture, Warranty registration
- **Generic** — Bare cart, no extras

How to switch a domain pack

1. Go to /app/alphax-pos-outlet
2. Open the outlet
3. **Domain Pack** field: pick the right one
4. Save

Cashiers at this outlet will see the relevant features enabled on next page reload.

Mixing packs across outlets

Each outlet has its own pack, even within a single branch. So a “Riyadh Mall” branch can have both “Coffee Shop” (Restaurant pack) and “Pharmacy” (Pharmacy pack) outlets, each with the right features.

Loyalty, offers, pricing rules

✓ Working

Loyalty program setup

1. Go to /app/loyalty-program/new
2. Define earn rate (e.g., 1 point per 1 SAR spent)
3. Define redemption value (e.g., 100 points = 10 SAR off)
4. Set expiry policy (typical: 1 year)
5. Save

In POS Profile, link the Loyalty Program. Cashiers can see and apply loyalty automatically.

Offers (campaigns)

1. Go to /app/alphax-pos-offer/new

2. Code: a coupon code customers can type at checkout
3. Discount: percent or amount
4. Date range
5. Item / Item Group filter
6. Min purchase
7. Save

The cashier's "Offer Code" field accepts these codes.

Pricing rules

For non-coupon discounts (volume discounts, Tuesday discounts, etc.), use ERPNext's standard Pricing Rule doctype. See Day 4 above.

Pharmacy domain: drugs, prescriptions, controlled substances

✅ **Working** for the data model 🧪 **Beta** for the cashier UI integration (Vue cashier only) 🚧
In Development — full prescription capture in Classic cashier

11.1 Drug Master setup

Each medication is an Item AND an AlphaX Drug Master record:

1. Item Code is the same on both
2. AlphaX Drug Master adds:
 - Active ingredient
 - Strength
 - Form (tablet, capsule, syrup, etc.)
 - Route (oral, topical, IV)
 - Schedule (Schedule II, IV, OTC)
 - Max refills, refill cap

11.2 Drug Interaction Rules

Define which combinations are dangerous:

1. Go to `/app/alphax-drug-interaction-rule/new`
2. Drug A, Drug B
3. Severity: Major / Moderate / Minor
4. Mechanism (free-text explanation)
5. Save

The cashier will warn at scan time when a customer tries to buy two interacting drugs in the same transaction.

11.3 Controlled Substance Log

Every dispense of a Schedule II/III/IV drug auto-creates an AlphaX Controlled Substance Log entry. This is a regulatory requirement in some jurisdictions.

The log has: - Drug, quantity, batch - Patient identifier - Prescribing physician - Pharmacist who dispensed - Timestamp

Auditable. Cannot be deleted by users (System Manager too).

11.4 Prescription capture

 **In Development** — for v15.5.13, prescriptions are entered as documents (/app/alphax-prescription/new) and the cashier sale is linked to them via the customer field. Full inline prescription entry from the cashier is on the roadmap.

Routine administration tasks

12.1 Daily

- Check ZATCA Status (workspace shortcut) for any failed submissions
- Review Manager Authorization Log for unusual patterns
- Confirm all shifts closed cleanly yesterday

12.2 Weekly

- Review Sales Register for variance trends
- Check inventory adjustments
- Backup verification (see “Backup and disaster recovery” below)

12.3 Monthly

- Rotate manager PINs (optional but recommended)
- Review Item pricing — anything stale?
- ZATCA certificate expiry check (most renew yearly)
- User access audit — anyone left the company?

12.4 Yearly

- ZATCA recertification (per ZATCA’s schedule)
 - Frappe Cloud bench upgrade (Frappe v15 – v16 when stable)
 - Customer’s tax-year close
-

Backup and disaster recovery

 **Frappe Cloud handles this** for you.

What Frappe Cloud does automatically

- Daily SQL dumps of the site database

- Daily file backups (uploaded files, certificates, etc.)
- 14 days of restore points kept

What you should do additionally

- **Monthly off-site copy:** download a backup to your own storage monthly. Frappe Cloud is reliable but a customer-side copy is defense in depth.
- **Test a restore:** at least once, verify a backup can actually restore. Don't discover backup is broken in an emergency.

Disaster scenarios

Site is unreachable: contact Frappe Cloud support. Usually resolved in <1 hour.

Database corruption: restore from latest backup. You lose any sales between backup and corruption (typically <24 hours). Use the offline Sync Queue on cashier PCs if any sales weren't synced when corruption happened.

Customer's bench is decommissioned: download all backups, save them. Customer can re-deploy on a new bench from these backups.

Troubleshooting

Cashier shows "Station Not Configured" forever

Manager hasn't bound the PC yet. See Day 5 — "Test PIN authentication".

Manager PIN dialog says "Incorrect credentials"

Possible causes: - Manager mistyped → try again - Manager doesn't have the AlphaX POS Manager role → grant it - PIN was never set for this manager → set it (Day 5.4) - Manager is locked out → reset (Day 5.7)

The error is intentionally generic to prevent enumeration of valid managers. Look at the **AlphaX POS Manager Authorization Log** to see what actually happened.

Some assets serve, others 404

If `/assets/alphax_pos_suite/dist/vendor/...` works but `/assets/alphax_pos_suite/dist/cashier/...` returns 404 — this is a Frappe Cloud bench issue. Run from the bench shell:

```
bench --site <your-site> build --app alphax_pos_suite --hard-link
bench --site <your-site> migrate
bench restart
```

If still failing, contact Frappe Cloud support — the symlink for the app's public folder may be broken.

ZATCA invoice failed

Drill into the row in **AlphaX POS Processing Log**. The error column tells you why. Common fixes:

- “Invalid VAT number” → fix on the Customer record, resubmit
- “Device certificate expired” → reissue via the ZATCA companion app
- “Time skew” → check NTP on bench server (Frappe Cloud does this automatically; report to support if persistent)

Slow cashier on a specific PC

Possible causes: - Network: check ping to your bench - Browser: try Chrome instead of older browsers - Hardware: bridge daemon eating CPU? Restart it - Cashier UI: hard-refresh (Ctrl+Shift+R) to clear cached JS

Audit log is empty even after manager actions

Check that `frappe.log_error` isn't suppressing failures. From the bench console:

```
import frappe
frappe.get_all("Error Log", fields=["error", "creation"], limit=5)
```

If you see “AlphaX POS: failed to write authorization audit log” errors, the audit doctype itself may have issues. Run migrate, then re-test.

Glossary

Bench — A Frappe deployment unit. One bench can host multiple sites. Frappe Cloud customers usually have one bench per Frappe Cloud account.

Bcrypt — A password-hashing algorithm. Used for manager PINs. Slow by design (12 rounds takes ~50ms) — defends against offline brute-force.

Branch — A standard Frappe doctype for physical store locations.

Bridge — The AlphaX POS Bridge daemon, a separate Python program running on each cashier PC. Handles USB hardware.

CSID — Cryptographic Stamp Identifier. The certificate ZATCA issues per device for Phase 2 e-invoicing.

Domain Pack — A bundle of features turned on/off for a vertical (Restaurant, Retail, Pharmacy, etc.). Set per outlet.

Fatoora — ZATCA's portal for VAT registration and e-invoicing device management.

Outlet — An AlphaX-specific doctype for business units within a branch.

POS Profile — Standard ERPNext doctype defining cashier permissions and settings.

Sandbox / Production (ZATCA) — Two ZATCA environments. Sandbox for testing, Production for live invoices.

System Manager — Frappe role with full access. Use sparingly. Required for: bench-level operations, setting manager PINs, clearing lockouts, accessing audit logs at scale.

Terminal — An AlphaX-specific doctype for individual cashier stations within an outlet.

Quick reference: cheat sheet

Task	Where
New install	Bench → Apps → Add App from GitHub
Create a Branch	/app/branch/new
Create an Outlet	/app/alphax-pos-outlet/new
Create a Terminal	/app/alphax-pos-terminal/new
Create POS Profile	/app/pos-profile/new
Set Manager PIN	Bench → Console → <code>frappe.call(...)</code>
ZATCA Status	/app/alphax-pos-processing-log
Audit Log	/app/alphax-pos-manager-authorization-log
User management	/app/user
Today's Sales	/app/sales-register (filtered today)
Workspace	/app/alphax-pos-hub

Manual version 15.5.13 — May 2026.